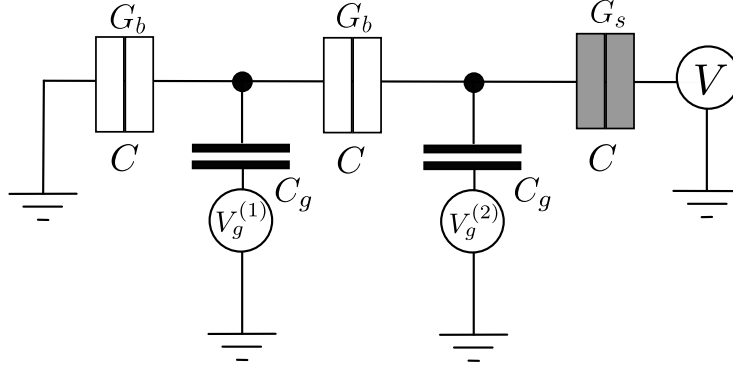


9 Task #9. Coulomb staircase



We study the transport in the three-junction circuit given in the Figure. If $G_b \gg G_s$, the $I - V$ curve exhibits a typical form called Coulomb staircase. At sufficiently low temperature each step starts with a cusp where differential conductance jumps from low to high value. At further increase of voltage, the differential conductance changes rather smoothly to low value and stays at this value till the next step. We set $C = 1.0$, $G_s = 1.0$, $C_g = 0.01$ for all numerical calculations.

- (2 points) Let us set $T = 10^{-3}$, $V_g^{(1)}, V_g^{(2)} = 0$. Make plots of $I - V$ curves with a relative accuracy at least 10^2 for a set of different $G_b = 50, 100, 200, 400$. Verify that the step positions do not depend on G_b . Give the high value of differential conductance for each step and each value of G_b . Suggest a scaling formula for the high conductance as function of G_b .
- (3 points) Let us set $G_b = 100$ and study the dependence of step positions on the charges induced in the two islands, $q_{1,2}$. Let us fix $q_1 = 0, G_b = 100.0$. Investigate the dependence of the following quantities on q_2 : step position, high conductance, low conductance for the steps 1, 2, 3; for at least 30 points in the interval $q_2 = (-0.5, 0.5)$. Bring all data in a single plot. Repeat all this for $q_1 = 0.5$.
- (3 points) Let us study temperature dependence of the $I - V$ curves. Set $G_b = 100$, $q_{1,2} = 0$. Concentrate on the start of the first step and compute at least 10 curves at different temperatures in the interval $T = (10^{-4}, 10^{-2})$. Suggest a parameter to characterize the rounding of the cusp and a scaling formula for this temperature-induced rounding. Make plots in wider voltage interval encompassing 3 steps to establish at which temperature the staircase structure is not visible anymore.
- (2 points) The start of each step corresponds to a new charging state that starts to participate in transport. Consider the conditions of point b. Use the tool **CBSetup.sorted_cache()** before and after the start of the step to identify this state for each of the three steps and all values of q_2 .